Capacity-Radius Proximity for Smooth-Domain Reed–Solomon Codes

Chloe van der Vlugt

Abstract

We study list decoding and mutual correlated agreement for constant-rate Reed–Solomon codes over smooth evaluation domains. The primary received-word space is the two-fold interleaving $\mathcal{C}^{=2} \subseteq (\mathbb{F}^2)^n$. We isolate three layers that are often conflated: normalized distance of a received word from the interleaved code, worst-case list size at that radius, and the fraction of affine challenges for which a scalar projection appears close while its underlying pair does not. A field-valued two-coordinate plane is exactly an arbitrary word in $(\mathbb{F}^2)^n$, and it is equivalently the base–direction pair of an affine line of scalar received words. This gives a lossless interface between interleaved list decoding and mutual correlated agreement.

The target regime is a strict finite approximation to the list-decoding capacity radius $1-\rho$, with explicit comparison against a security target $\varepsilon^*|\mathbb{F}|$. At the present draft stage, the capacity-radius list-size and mutual-correlated-agreement estimates are stated as proof targets rather than completed theorems. This separation is intentional: representation and metric equivalences do not by themselves establish the required quantitative coding bounds.

Contents

1 Introduction **2**
- 1.1 Central organization . . . 3
- 1.2 Claim discipline . . . 3
- 1.3 Contributions sought . . . 4

2 Problem Setup **4**
- 2.1 Reed–Solomon parameters . . . 4
- 2.2 Relative Hamming distance . . . 4
- 2.3 Interleaving . . . 5
- 2.4 List-decoding balls . . . 5

3 Planes, Pair Words, And Affine Lines **5**
3.1 The full ambient pair space 5
3.2 Pair words as affine lines 6
3.3 Challenge evaluation is lossy 6

4 The Capacity-Radius Regime **7**
4.1 Normalized target 7
4.2 Field-size scaling 7
4.3 Largest safe radius 8

5 The List-Decoding Target **8**
5.1 What a proof must control 8
5.2 Restricted and universal statements 9
5.3 Proof status 9

6 The Mutual-Correlated-Agreement Target **9**
6.1 Bad challenges 9
6.2 Distance interpretation 10
6.3 Commit-then-challenge order 10
6.4 Proof status 10

7 Application Interface **11**
7.1 Exact witnesses versus raw observations 11
7.2 Separation from application papers 11

8 Formalization And Proof Ledger **12**
8.1 Standalone hierarchy 12
8.2 Status ledger 12
8.3 Submission boundary 13
8.4 Immediate next step 13

1 Introduction

Reed–Solomon proximity questions connect three quantitatively different objects:

(i) the relative distance of a received word from a code;

(ii) the number of codewords in a ball at that radius;

(iii) the fraction of affine challenges that conceal distance of an underlying tuple of received words.

The Grand List Decoding and Grand Mutual Correlated Agreement challenges ask for sharp forms of the latter two quantities for constant-rate Reed–Solomon codes over smooth evaluation domains [ABF26].

This paper treats the coding problem independently of any protocol or application-specific source of received words. Its universal statements quantify over every word in the ambient alphabet and every affine line. A later application may map structured observations into this ambient space, but that map is not part of the coding theorem.

1.1 Central organization

Let

$$\mathcal{C} = \mathrm{RS}[\mathbb{F}, \mathcal{L}, k] \subseteq \mathbb{F}^n, \qquad n = |\mathcal{L}|, \qquad \rho = \frac{k}{n}.$$

The two-fold interleaving is

$$\mathcal{C}^{\equiv 2} \subseteq (\mathbb{F}^2)^n.$$

Every pair word $p \in (\mathbb{F}^2)^n$ determines an affine line

$$y_\gamma(p) = p_0 + \gamma p_1 \in \mathbb{F}^n, \qquad \gamma \in \mathbb{F}.$$

The intended proof architecture is therefore

$$\begin{array}{ccc} p \in (\mathbb{F}^2)^n & \xrightarrow{\gamma \in \mathbb{F}} & y_\gamma(p) \in \mathbb{F}^n \\ \text{distance and list size relative to } \mathcal{C}^{\equiv 2} & & \text{distance relative to } \mathcal{C}. \end{array}$$

List decoding controls ambiguity of the full pair; MCA controls the fraction of challenges for which a scalar observation fails to expose pair-farness.

1.2 Claim discipline

The plane/pair/line identifications below are exact equivalences. They ensure that no received words or affine lines are lost by the chosen presentation. They do not prove capacity-radius list decoding or MCA. Those quantitative statements are separately labeled as proof targets until a complete argument and its parameter inequalities have been supplied.

In particular, a statement of the form

$$\mathrm{dist}(p, \mathcal{C}^{\equiv 2}) \geq \delta$$

is not a list-size bound, and neither fact alone implies a security estimate of 2^{-128}. The required list size or bad-challenge count must be compared explicitly with $\varepsilon^* |\mathbb{F}|$.

1.3 Contributions sought

The completed paper is intended to establish:

(i) a capacity-radius list-size estimate for $\mathcal{C}^{\equiv m}$, with $m = 2$ as the primary plane endpoint;

(ii) a mutual-correlated-agreement estimate at the same normalized radius;

(iii) explicit field-size hypotheses sufficient to obtain a selected ε^*;

(iv) a machine-checked formalization of the definitions, reductions, and final quantitative theorem.

2 Problem Setup

2.1 Reed–Solomon parameters

Let $\mathbb{F}$ be a finite field and let $\mathcal{L} = (a_i)_{i \in I}$ be an injectively indexed evaluation domain, where $|I| = n$. For $1 \le k \le n$, define

$$\mathcal{C} = \mathrm{RS}[\mathbb{F}, \mathcal{L}, k] = \{(f(a_i))_{i \in I} : f \in \mathbb{F}[X],\ \deg f < k\}\,.$$

The rate is $\rho = k/n$. The prize-facing regime takes ρ from a fixed constant set, such as

$$\left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\right\},$$

and requires $\mathcal{L}$ to have an appropriate smooth structure.

2.2 Relative Hamming distance

For a finite alphabet A and $u, v \in A^I$, let

$$d_H(u, v) = |\{i \in I : u_i \neq v_i\}|, \qquad \Delta(u, v) = \frac{d_H(u, v)}{n}.$$

For a code $\mathcal{D} \subseteq A^I$, define

$$\Delta(w, \mathcal{D}) = \min_{c \in \mathcal{D}} \Delta(w, c).$$

All radii in the main theorem statements are relative radii in $[0, 1]$. Natural-number radii remain useful for finite rounding, but they are not the paper-facing metric.

2.3 Interleaving

For a positive integer m, define

$$\mathcal{C}^{\equiv m} = \left\{ p \in (\mathbb{F}^m)^I : \text{for every } j \in \{1, \ldots, m\},\ (p_{i,j})_{i \in I} \in \mathcal{C} \right\}.$$

Hamming distance on $(\mathbb{F}^m)^I$ counts a coordinate i as erroneous when the complete vector symbol differs. It does not sum componentwise scalar distances.

2.4 List-decoding balls

For $w \in (\mathbb{F}^m)^I$ and $\delta \in [0, 1]$, define

$$\Lambda(\mathcal{C}^{\equiv m}, w, \delta) = \{ c \in \mathcal{C}^{\equiv m} : \Delta(w, c) \leq \delta \}.$$

The worst-case list size is

$$L_m(\mathcal{C}, \delta) = \max_{w \in (\mathbb{F}^m)^I} |\Lambda(\mathcal{C}^{\equiv m}, w, \delta)|.$$

The maximum over received words is essential. A bound on a structured subset of received words is an application result, not the universal list-decoding statement.

3 Planes, Pair Words, And Affine Lines

3.1 The full ambient pair space

Define a field-valued two-coordinate plane to be a function

$$p : I \longrightarrow (\{0, 1\} \longrightarrow \mathbb{F}).$$

By reassociating function spaces, the plane carrier is exactly

$$I \to (\{0, 1\} \to \mathbb{F}) \cong (\mathbb{F}^2)^I.$$

Thus the word "plane" introduces no restriction on the received-word space.

Proposition 3.1 (Plane–pair equivalence)**.** *The map sending a plane p to the vector word*

$$i \longmapsto (p(i, 0), p(i, 1))$$

is a bijection from field-valued planes to $(\mathbb{F}^2)^I$. It preserves and reflects Hamming distance.

Proof. The inverse sends a pair word w to the function

$$(i, j) \longmapsto w_i(j).$$

Both composites are pointwise identities. At coordinate i, the two planes are equal exactly when their complete pair symbols are equal, so the Hamming disagreement sets coincide. □

3.2 Pair words as affine lines

For a plane p, define its base and direction by

$$p_0(i) = p(i, 0), \qquad p_1(i) = p(i, 1).$$

The associated affine line of scalar received words is

$$\ell_p(\gamma) = p_0 + \gamma p_1, \qquad \gamma \in \mathbb{F}.$$

Proposition 3.2 (Plane–line equivalence)**.** *The map $p \mapsto (p_0, p_1)$ is a bijection between field-valued planes and affine-line presentations consisting of an arbitrary base word and arbitrary direction word in $\mathbb{F}^I$.*

Proof. Given an arbitrary pair (f_0, f_1), define

$$p(i, 0) = f_0(i), \qquad p(i, 1) = f_1(i).$$

This construction is inverse to taking base and direction. □

Remark 3.3. *These equivalences discharge the coverage question: a theorem quantified over all planes is a theorem over every received word in $(\mathbb{F}^2)^I$ and every affine-line presentation. No surjectivity claim from an application-specific frame or commitment space is required.*

3.3 Challenge evaluation is lossy

For fixed γ, the map

$$(p_0, p_1) \longmapsto p_0 + \gamma p_1$$

is generally many-to-one. Consequently, pair-farness cannot be inferred deterministically from one fixed challenge without an additional good-challenge hypothesis. MCA supplies a probabilistic statement over a challenge sampled after the pair has been fixed.

4 The Capacity-Radius Regime

4.1 Normalized target

For rate $\rho = k/n$, the list-decoding capacity benchmark is

$$1 - \rho.$$

At finite block length we use an explicitly strict integer radius

$$r_{\mathrm{cap}} < n - k, \qquad \delta_{\mathrm{cap}} = \frac{r_{\mathrm{cap}}}{n} < 1 - \rho.$$

The strict inequality separates the intended capacity regime from endpoint rounding questions.

For the running rate-1/4, length-4096 example,

$$n = 4096, \qquad k = 1024, \qquad r_{\mathrm{cap}} = 3071,$$

and hence

$$\delta_{\mathrm{cap}} = \frac{3071}{4096} < \frac{3}{4}.$$

This example is a finite parameter checkpoint, not the sufficiently-large-field family required for a fixed-bit security claim.

4.2 Field-size scaling

Fix a target error fraction ε^*, such as 2^{-128}. A quantitative list or MCA theorem must produce a bound B satisfying

$$B \leq \varepsilon^* |\mathbb{F}|.$$

For an integer-valued nonempty list, this already forces $|\mathbb{F}| \geq 1/\varepsilon^*$. Therefore a small fixed field cannot instantiate the grand list-decoding target at cryptographic ε^*, regardless of metric normalization.

The main theorem must consequently be parameterized by a family

$$(\mathbb{F}_\lambda, \mathcal{L}_\lambda, n_\lambda, k_\lambda)$$

with constant rate, smooth evaluation domains, and field size growing sufficiently quickly relative to the derived list-size or bad-challenge bound.

4.3 Largest safe radius

For fixed $\mathcal{C}, m, \varepsilon^*$, define the list-decoding safe-radius set

$$\mathcal{R}_{\text{list}} = \{\delta \in [0,1] : L_m(\mathcal{C}, \delta) \le \varepsilon^* |\mathbb{F}|\}.$$

A grand-challenge solution must not merely exhibit one member of this set; it must determine or tightly characterize its largest element. The first proof milestone in this manuscript is more modest: establish that a strict capacity-approaching radius belongs to this set under explicit field-size hypotheses.

5 The List-Decoding Target

Problem 5.1 (Capacity-radius interleaved list bound)**.** *Let $\mathcal{C} = \mathrm{RS}[\mathbb{F}, \mathcal{L}, k]$ have constant rate and a smooth evaluation domain. For fixed m, determine an explicit bound*

$$L_{\text{cap}}(\mathbb{F}, n, k, m)$$

such that

$$\forall w \in (\mathbb{F}^m)^n, \qquad |\Lambda(\mathcal{C}^{\equiv m}, w, \delta_{\text{cap}})| \le L_{\text{cap}}(\mathbb{F}, n, k, m).$$

Determine field-size conditions under which

$$L_{\text{cap}}(\mathbb{F}, n, k, m) \le \varepsilon^* |\mathbb{F}|.$$

For the plane endpoint, $m = 2$. By Theorem 3.1, the universal quantifier may equivalently be written over every field-valued plane. This is a change of presentation, not a restriction to a structured image.

5.1 What a proof must control

A complete argument must specify:

(i) the exact strict capacity radius and its finite rounding;

(ii) the dependence of list size on n, k, m, and $|\mathbb{F}|$;

(iii) the use made of smoothness of the evaluation domain;

(iv) whether the result holds for all such domains or a selected family;

(v) the field-size threshold needed for the target ε^*;

(vi) the comparison with Johnson-radius and known proximity-gap results.

5.2 Restricted and universal statements

For a subset $Y \subseteq (\mathbb{F}^m)^n$, the statement

$$\forall w \in Y, \qquad |\Lambda(\mathcal{C}^{\equiv m}, w, \delta)| \leq L$$

is a restricted-image list bound. It may be useful for an application, but it does not imply Theorem 5.1 unless every received word reduces to an element of Y while preserving the relevant list.

The plane carrier avoids this issue because $Y = (\mathbb{F}^2)^n$. An upstream application image, such as words produced by valid geometric witnesses, need not and generally will not have the same coverage.

5.3 Proof status

[TODO: Insert the capacity-radius list-size argument.]

Until that argument is supplied, no theorem in this paper should assert that δ_{cap} belongs to $\mathcal{R}_{\text{list}}$.

6 The Mutual-Correlated-Agreement Target

6.1 Bad challenges

Let $p = (p_0, p_1) \in (\mathbb{F}^2)^I$, and write

$$y_\gamma = p_0 + \gamma p_1.$$

For $S \subseteq I$, say that S is δ-large when

$$\frac{|S|}{n} \geq 1 - \delta.$$

Define $\text{Bad}_{\text{MCA}}(\mathcal{C}, \delta, p)$ to be the set of $\gamma \in \mathbb{F}$ for which there exists a δ-large set S such that:

(i) y_γ agrees on S with a codeword of $\mathcal{C}$;

(ii) the pair p does not agree on the same S with a codeword of $\mathcal{C}^{\equiv 2}$.

The MCA error is

$$\varepsilon_{\text{MCA}}(\mathcal{C}, \delta) = \max_{p \in (\mathbb{F}^2)^I} \frac{|\,\text{Bad}_{\text{MCA}}(\mathcal{C}, \delta, p)|}{|\mathbb{F}|}.$$

Problem 6.1 (Capacity-radius MCA bound)**.** *For a constant-rate smooth-domain Reed–Solomon code, determine an explicit bound*

$$B_{\mathrm{MCA}}(\mathbb{F}, n, k)$$

such that

$$\forall p \in (\mathbb{F}^2)^n, \qquad |\operatorname{Bad}_{\mathrm{MCA}}(\mathcal{C}, \delta_{\mathrm{cap}}, p)| \le B_{\mathrm{MCA}}(\mathbb{F}, n, k),$$

and determine when

$$\frac{B_{\mathrm{MCA}}(\mathbb{F}, n, k)}{|\mathbb{F}|} \le \varepsilon^*.$$

6.2 Distance interpretation

Let

$$D_2(p) = \Delta(p, \mathcal{C}^{\equiv 2}), \qquad D_1(y) = \Delta(y, \mathcal{C}).$$

For a challenge outside the MCA bad set, scalar closeness on a large agreement set forces pair closeness on that same set. In distance language, the desired contrapositive has the form

$$D_2(p) \ge \delta \Longrightarrow D_1(y_\gamma) \ge \delta$$

for every MCA-good γ, with the fraction of excluded challenges bounded by $\varepsilon_{\mathrm{MCA}}(\mathcal{C}, \delta)$.

This is a challenge-indexed statement. Replacing the sampled challenge by one fixed field element would require a separate theorem that the selected element is good for every pair, which does not follow from a fractional MCA bound.

6.3 Commit-then-challenge order

The mathematical experiment fixes p before sampling $\gamma \leftarrow \mathbb{F}$. This ordering prevents the pair from being selected as a function of the challenge. A full protocol implementation must bind p before deriving γ, but the coding theorem needs only the resulting quantification and probability space; it does not depend on a complete blockchain or commitment protocol.

6.4 Proof status

[TODO: Insert the capacity-radius MCA argument and its explicit field-size bound.]

7 Application Interface

The coding theorem is stated over the complete received-word space and does not depend on a particular source of observations. An application may nevertheless inherit the result through a distance-aware map.

Let O be a type of raw presented observations, let $G \subseteq O$ be the exact/valid locus, and let

$$\pi : O \longrightarrow (\mathbb{F}^2)^I$$

be the extracted pair word. A source distance $\rho_O : O \to [0,1]$ is meaningful only when O includes arbitrary or noisy presentations; if every element of O already contains a validity witness, then the proximity problem may collapse to the exact code locus.

The principal application theorem should have one of the forms

$$D_2(\pi(o)) = \rho_O(o),$$

or

$$\rho_O(o) \geq \delta \Longrightarrow D_2(\pi(o)) \geq \delta.$$

Under either comparison, the universal list and MCA theorems apply to $\pi(o)$.

7.1 Exact witnesses versus raw observations

The recommended type boundary is:

raw observation	arbitrary public presentation,
exact witness	certificate of valid structured realization,
$G(o)$	there exists an exact witness realizing o.

An exact witness is evidence for membership in the code locus; it should not be the carrier of adversarial received words.

7.2 Separation from application papers

Application-specific Plucker coordinates, cochain commitments, hidden operators, transcript binding, and runtime verification are intentionally outside the proof of Theorems 5.1 and 6.1. They may cite the completed coding theorem through the interface above. Conversely, the coding theorem must remain valid without assuming that every received word comes from an application-specific witness space.

8 Formalization And Proof Ledger

8.1 Standalone hierarchy

The intended machine-checked development should separate generic coding mathematics from application bridges:

`ProximityPrize/BlockCode`
: Relative distance, balls, and worst-case list size.

`ProximityPrize/ReedSolomon`
: Smooth-domain RS families and interleaving.

`ProximityPrize/PlanePair`
: Plane–pair–line equivalences.

`ProximityPrize/Parameters`
: Explicit ε^* and field-size inequalities.

Existing generic definitions may be imported or refactored, but the final theorem should not depend on HCP, protocol, consensus, or runtime axiom bundles. The open capacity-list and capacity-MCA targets are intentionally not declared as Lean theorems or axioms at this stage.

8.2 Status ledger

Obligation	Status	Required endpoint
Relative Hamming normalization	Lean-defined	Values in $[0, 1]$
Plane–pair equivalence	Lean-proved	Bijection and Hamming isometry
Plane–affine-line equivalence	Lean-proved	Formal inverse maps
Smooth large-field family	Open	Explicit domain family
Capacity-radius list bound	Open	Universal list-size theorem
Capacity-radius MCA bound	Open	Universal bad-set theorem
Target-fraction inequality	Open	Bound $\leq \varepsilon^*\lvert\mathbb{F}\rvert$
Largest-radius or optimality statement	Open	Safe-radius characterization

8.3 Submission boundary

A completed claim requires:

(i) no placeholder assumptions for the main list/MCA result;

(ii) an explicit statement of imported literature theorems;

(iii) all field-size and rounding conditions checked;

(iv) a clear separation between proved theorems, computational evidence, and conjectures;

(v) a reproducible proof artifact and manuscript cross-reference.

8.4 Immediate next step

The next mathematical task is to choose the exact capacity-approaching theorem statement for $m = 2$ and a parameterized smooth-domain field family. Only after those choices are fixed should the Lean target signatures and the core list-size argument be stabilized.

References

[ABF26] Gal Arnon, Dan Boneh, and Giacomo Fenzi. Open problems in list decoding and correlated agreement, 2026. July 2026 revision.